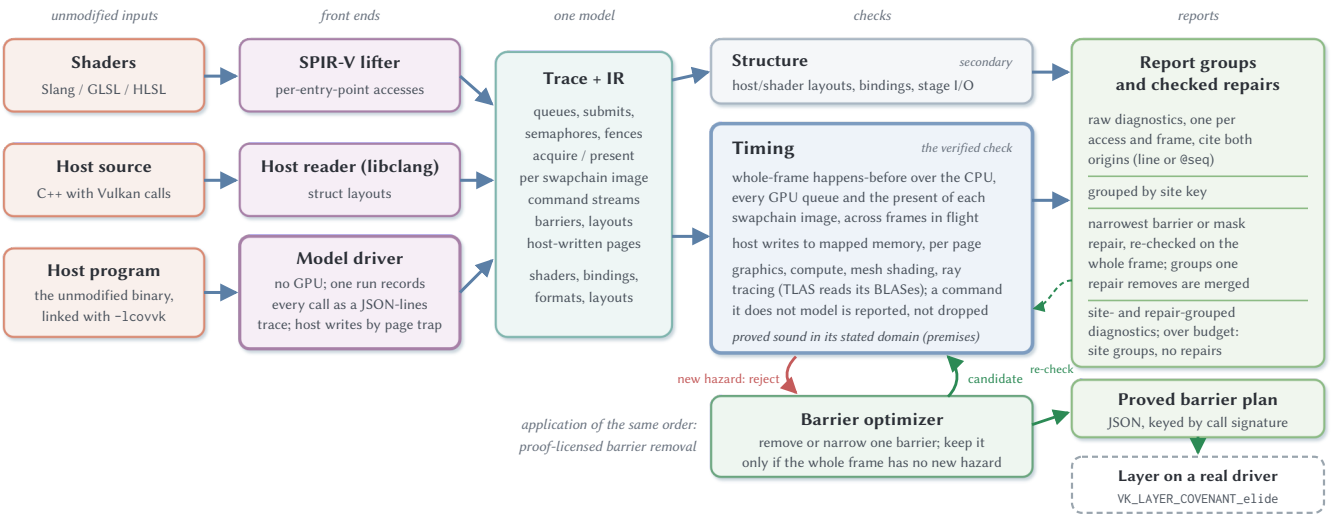

unmodified inputs
front ends
one model
checks
reports
Shaders
Slang / GLSL / HLSL
SPIR-V lifter
per-entry-point accesses
Host source
C++ with Vulkan calls
Host reader (libclang)
struct layouts
Host program
the unmodified binary, linked with -lcovvk
Model driver
no GPU; one run records every call as a JSON-lines trace; host writes by page trap
Trace + IR
queues, submits, semaphores, fences
acquire / present
per swapchain image
command streams
barriers, layouts
host-written pages
shaders, bindings, formats, layouts
Structure
secondary
host/shader layouts, bindings, stage I/O
Timing
the verified check
whole-frame happens-before over the CPU, every GPU queue and the present of each swapchain image, across frames in flight
host writes to mapped memory, per page
graphics, compute, mesh shading, ray tracing (TLAS reads its BLASes); a command it does not model is reported, not dropped
proved sound in its stated domain (premises)
Report groups and checked repairs
raw diagnostics, one per access and frame, cite both origins (line or @seq)
grouped by site key
narrowest barrier or mask repair, re-checked on the whole frame; groups one repair removes are merged
site- and repair-grouped diagnostics; over budget: site groups, no repairs
new hazard: reject
candidate
re-check
application of the same order: proof-licensed barrier removal
Barrier optimizer
remove or narrow one barrier; keep it only if the whole frame has no new hazard
Proved barrier plan
JSON, keyed by call signature
Layer on a real driver
VK_LAYER_COVENANT_elide